



Understanding Price Variability and Key Determinants in Used Range Rover Evoque Market

Executive Summary

Stratstone Land Rover in Reading undertook this pilot study to better understand the pricing dynamics of used Range Rover Evoque vehicles. Using data collected from Auto Trader UK, the study examined both local market listings within a 30-mile radius of the dealership and national trends, with a total of 110 vehicles sampled. To ensure a representative analysis, stratified random sampling was applied by body type, giving adequate coverage to less common vehicle types while reflecting the dominance of SUVs in the population.

Population insights revealed that diesel SUVs dominate the market, with petrol and hybrid models being less common. Price comparisons showed variation across fuel types, gearbox, and number of past owners as expected.

While several factors influence pricing, mileage and gearbox type are the strongest and most consistent predictors. Higher mileage consistently reduced vehicle value with each mile £0.30 price decreased which is highly significant, while manual gearboxes were priced lower than automatics.

Fuel type and past owners showed differences in average prices (ANOVA), but these effects were less pronounced in regression analysis. Market movement over time was small but measurable, with an average price drop of ~£277 between October and November.

Price variability differed across categories. Hybrids and Petrol car owners showed the widest confidence intervals, making their prices less predictable, while diesel SUVs and vehicles with 1 to 4 past owners were more stable. Visualizing price vs. variability (e.g., a comparison matrix) can help the dealership prioritize categories that are both valuable and consistent, while identifying categories with high uncertainty that might need targeted attention.

The study provides actionable insights for inventory acquisition, pricing strategies, and marketing positioning, while establishing a repeatable, data-driven methodology for ongoing analysis.

Limitations include reliance on listed prices, potential human error in data collection, inability to snapshot closing prices and the fact that findings indicate correlations rather than definitive causal relationships.

Future reports could expand the dataset with additional vehicle features and collect data over multiple time points to better capture market trends. Using automated data collection and enhanced visualizations would improve accuracy and make insights easier to interpret.

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Introduction on Case Study

Stratstone Land Rover, Reading [postcode RG2 0QX] is a used Land Rover seller/dealer in Reading, UK. Approved by the manufacturer themselves, their range of used Land Rover vehicles are delivered with a host of benefits that guarantee customers are investing in a quality example.

To maintain its competitive edge and uphold its promise of value to customers, the dealership has initiated a data-driven pilot study to analyse the pricing dynamics of one of its most popular Land Rover models: the **Range Rover Evoque**.

The primary objective is to perform a comparative price analysis of used Land Rover Evoques, examining determinants of asking prices within a **30-mile radius** of the dealership and benchmarking these against national trends on Auto Trader. The insights generated will inform strategic decisions regarding inventory acquisition, pricing strategy, and marketing positioning.

As a pilot, this study is designed not only to deliver immediate actionable findings but also to establish a robust, repeatable methodology for ongoing analysis, with a clear pathway for feedback and iterative improvement. The company hopes to generate the report on a regular basis.

Objective and Scope

The study aims to –

- Establish national benchmark price ranges and market composition for the used Range Rover Evoque on Auto Trader.
- Identify and quantify the key determinants of asking prices for used Evoques within a 30-mile radius of the Stratstone Reading dealership.
- Develop a multivariate pricing model to predict vehicle asking prices based on its key features.

Sampling Methodology:

To ensure a comprehensive and representative analysis, a structured approach was employed.

Data Source: Auto Trader UK; URL: <https://www.autotrader.co.uk/>

Primary/Local Market: 30-mile radius of RG2 0QX [post code of Stratstone Land Rover car dealer]

Benchmark/National Market: National (UK-wide) data

Please note that, for both the Local and National market, population data was collected to understand the price distribution of the used Range Rover Evoque cars and to do a high-level comparison between the two markets.

Car Features and Information: Variables were selected based on availability from the Auto Trader website, prioritizing data points that could be consistently collected within the project's timeframe. The following variables were collected:

Upload Date, Reg Year, Price October 16, Mileage, Car Age (years), Engine size, Seats, Door, Acceleration (0-60mph), Cylinder, Valves, Engine Power (bhp), Fuel Consumption (mpg), Past Owners, Colour, GearBox, Body Type, Emmission Class, Fuel, Price November 05, Sold

Car Age (years) and Sold variables were calculated.

Sampling Method: A **Stratified Random** sampling method was applied to the local market vehicle listings. An initial population study revealed that diesel SUVs dominate the market. To prevent this majority group from overshadowing less common vehicle types in our sample, the population was stratified by **car body type** before sampling. This ensured a representative sample that captures the full spectrum of the available variables.

The table below outlines the data point distribution for the population and the resulting sample.

Body Type	Number of Listings	Percentage	Sampled	Sampled Percentage
Convertible	46	1%	2	2%
Coupe	117	3%	5	5%
Hatchback	1	0%	1	1%
SUV	4369	96%	102	93%

Consequently, the sample percentages do not perfectly match the population distribution. This intentional mismatch ensures that less common vehicle types (e.g., non-SUV or petrol/hybrid models) are adequately represented for analysis.

Total Data Points Collected: 110

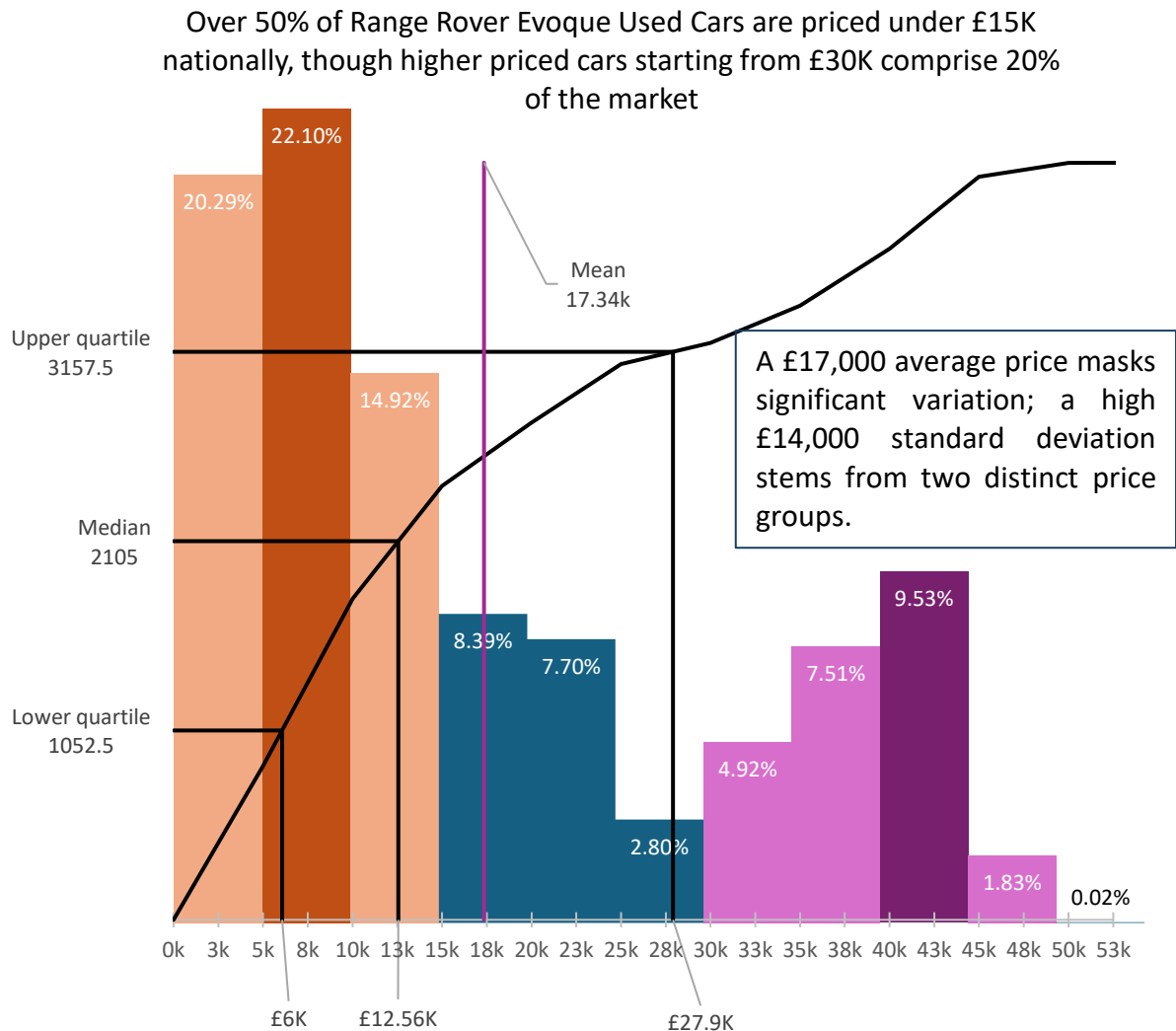
Limitations

- Manual Data Collection: Introduces risk of human error.

- Convenience-Based Variables: Limited to readily available data, omitting key factors like service history.
- Representation Bias: Stratified sampling may under-represent the most common vehicle types.
- Asking Price Focus: Analysis reflects listed prices, not final transaction values.
- Static Snapshot: Captures market conditions at a single point in time, missing price fluctuations.
- Correlation, Not Causation: Identifies relationships but cannot prove definitive cause-and-effect.

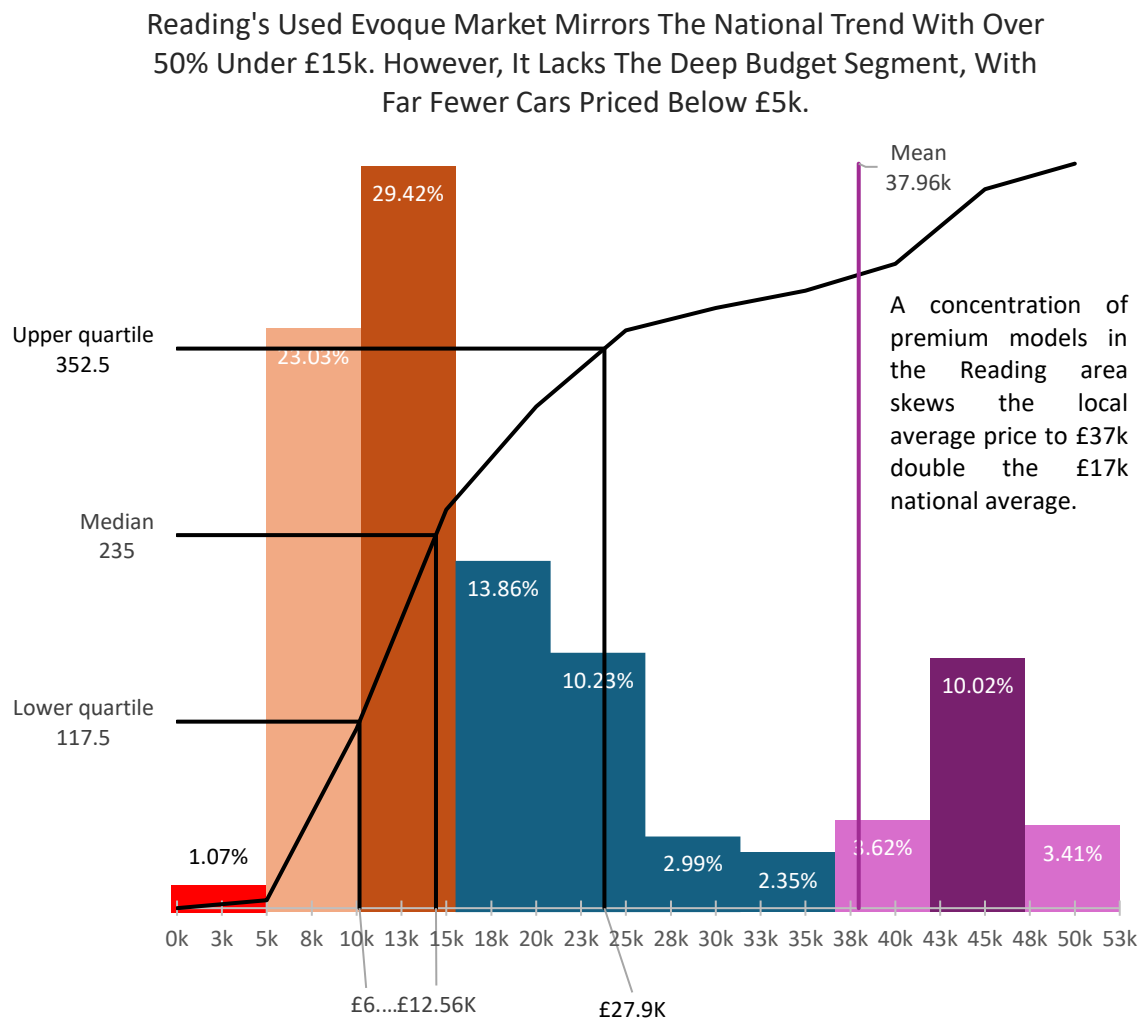
Case Study of Range Rover Evoque Used Cars

1.1 UNDERSTANDING POPULATION DISTRIBUTION: NATIONAL VS LOCAL



The chart adheres to all six principles of graphical integrity through proportional representation, clear labelling, and appropriate contextual information. It follows Tufte's principles of graphical excellence by presenting multivariate data with minimal ink-to-information ratio, efficiently conveying the complete pricing distribution of used cars. Through a carefully chosen color gradient (Colin Ware's per-attentive visual attribute), the visualization reveals the slight dual modality in price distribution while maintaining Gestalt principles of visual perception. The Institute of Analytics (IoA)'s guidance has also been

followed by applying insightful chart title that focuses on message that that chart is trying to provide to the viewers.



This chart follows the same structure as the previous one. Only the data set changed. Keeping in mind that similar chart helps viewer grasp messages faster and with more clarity.

1.2 VISUALIZING CONTINUOUS VARIABLES THAT PLAY CRUCIAL ROLE IN PRICING

1.2.1.1 Correlation Matrix and its Interpretation

Car Features Show Strong Correlations: Mileage, Age, and Year highly interrelated;

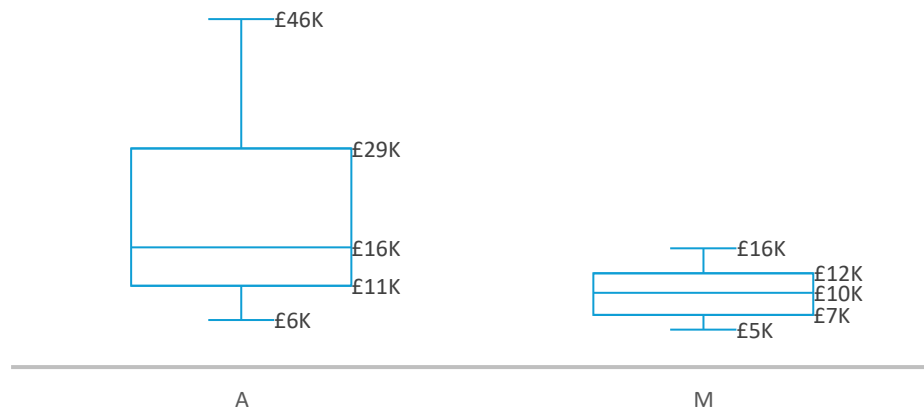
Cylinder, Engine Size, and Past Owners moderately correlated

	Price October 16	Mileage	Car Age (years)	Engine size	Seats	Door	Acceleration (0-60mph)	Cylinder	Valves	Engine Power (bhp)	Fuel Consumption (mpg)	Past Owners
Price (Oct 16)	1.0											
Mileage	-0.9	1.0										
Car Age (years)	-0.9	0.9	1.0									
Engine size	-0.4	0.4	0.5	1.0								
Seats	0.0	0.0	0.0	0.0	1.0							
Door	0.0	0.0	-0.1	-0.1	-0.5	1.0						
Acceleration (0-60mph)	-0.4	0.3	0.4	0.6	0.0	0.2	1.0					
Cylinder	-0.3	0.2	0.3	0.9	0.0	0.0	0.6	1.0				
Valves	-0.1	0.2	0.1	0.4	0.2	0.1	0.3	0.5	1.0			
Engine Power (bhp)	0.3	-0.3	-0.2	-0.5	-0.3	0.2	-0.6	-0.5	-0.3	1.0		
Fuel Consumption (mpg)	0.1	-0.1	-0.2	-0.6	-0.3	0.4	-0.3	-0.7	-0.3	0.6	1.0	
Past Owners	-0.5	0.5	0.6	0.4	-0.1	-0.1	0.2	0.2	0.2	-0.2	0.0	1.0

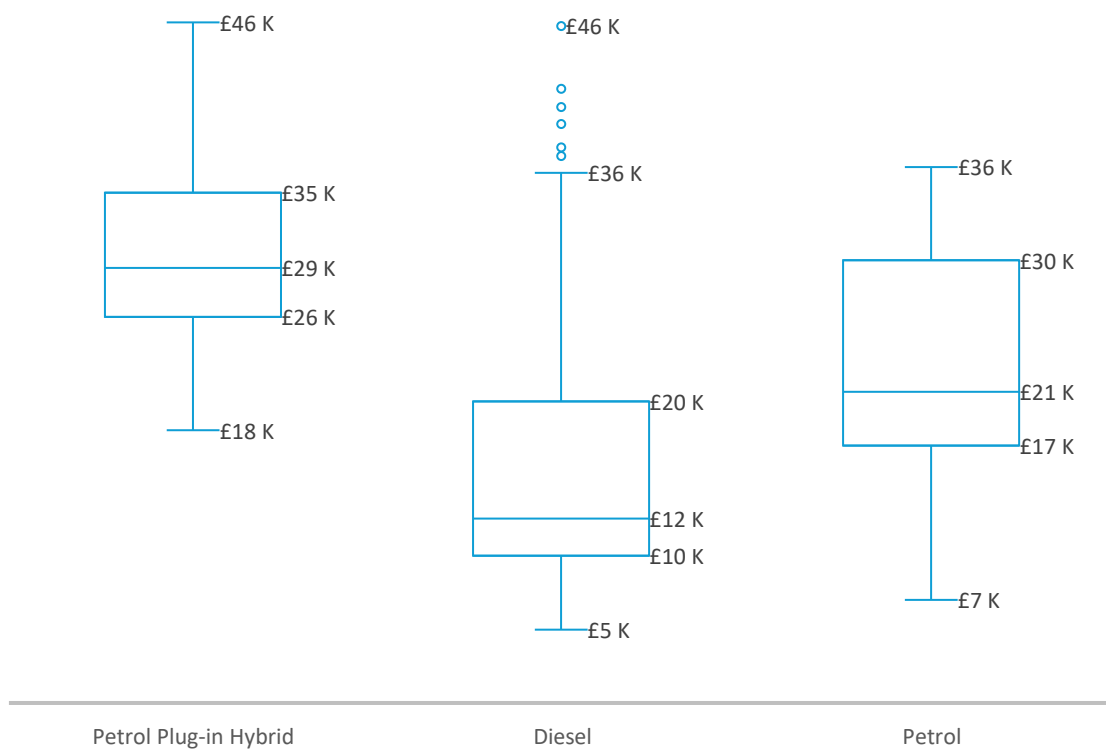
The correlation table here is designed to meet Cole's Imperative to Cut Clutter. There is no unnecessary grid lines, consistent colour has been used to identify at a glance which features are highly correlated. Bertin's encoding about using changes in hue to show intensity or direction of message is also incorporated here. The titles are boxed to give visual clarity and move the focus to the centre data area - which in a way follows Gestalt Principle 'Closure'.

1.3 VISUALIZING CATEGORICAL VARIABLES THAT PLAY CRUCIAL ROLE IN PRICING

Whilst Cars With **Automatic** Gear Have A Wide Range Of Price Points, **Manual** Cars Are Generally Cheaper And Do Not Have Higher Price Options



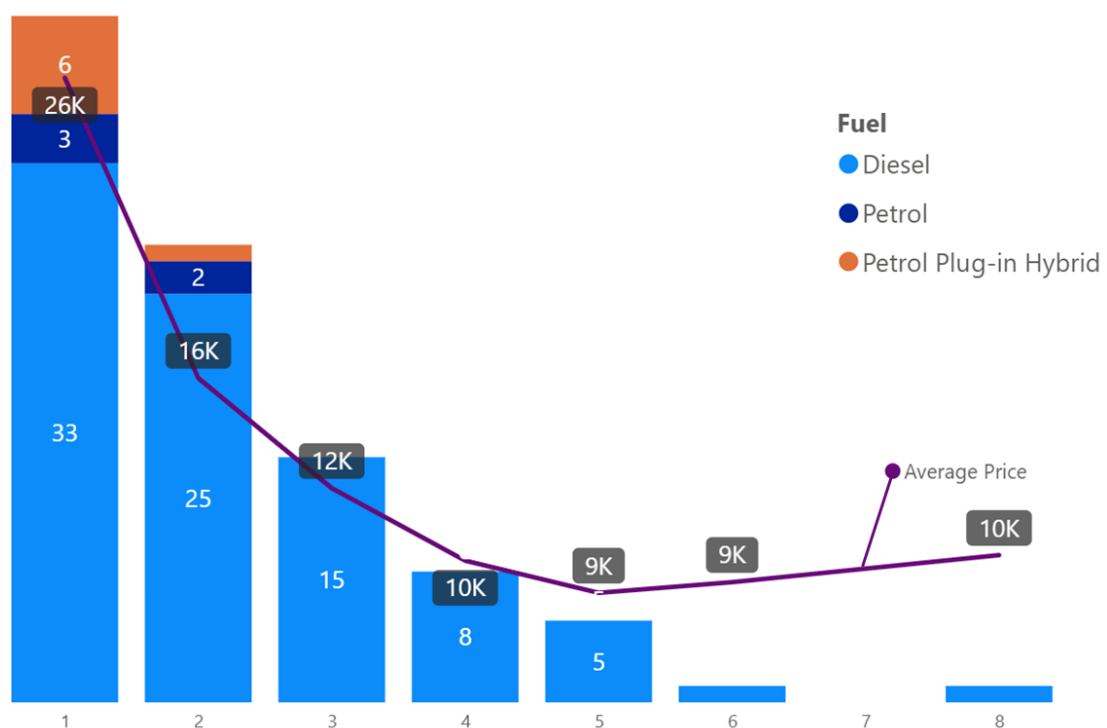
Hybrid Cars Start Off At A Higher Price Range, With An Average Of £30K. **Diesel Cars** Have The Most Versatile Price Offerings. **Petrol Cars** Fall Between Diesel And Hybrid, But Still Occupy The Upper Price Range



The usage of the simple box charts provides a holistic information about the price distribution of certain variables quickly without any distraction. Labels have been shortened to further emphasize simplicity – incorporating ‘more info-less ink’ principle.

Number of Previous Owners **Negatively Correlates** with Average Price of Range Rover

Evoque; Most Cars, Including Hybrids, Have Had **No More Than Two Owners**



The chart depicts the distribution of cars according to ownership and fuel type with clear labels, proportional scales, and relevant context. Its color gradient highlights subtle patterns, while the concise title and design make the key insights immediately understandable. Additionally it is clear that ownership decreases the Resale value.

1.4 STATISTICAL ANALYSIS AND ITS INTERPRETATION

1.4.1 Research Questions Hypothesis

Based on the correlation found between some of the variables and the price (collected on October 16), a number of research question hypotheses can be formulated -

Question 1: Does average price (collected on October 16) differ by Fuel type, Gear type, Mileage or Past Owners? So the hypothesis tests will be:

H0: All group means (by Fuel type, Gear type, Mileage or Past Owners) are equal

H1: At least one group mean differs

Question 2: Is there a change between the price on October 16 and November 5?

H0: The mean difference between the prices are 0

H1: The prices are different

1.4.1.1 *One-way ANOVA Results for Price Addressing Question 1: Mean price vary by any group type*

Variable (Factor)	df (Between, Within)	F-Statistic	p-Value	Significance ($\alpha=0.05$)	Interpretation
Fuel Type	(2, 97)	4.28	0.0165	Significant	Price differs significantly by fuel type; hybrids and petrol cars tend to have higher mean prices than diesel.
Gear Box	(1, 98)	13.48	0.00039	Highly Significant	Automatic models are priced higher than manual ones, showing a clear market preference and value difference.

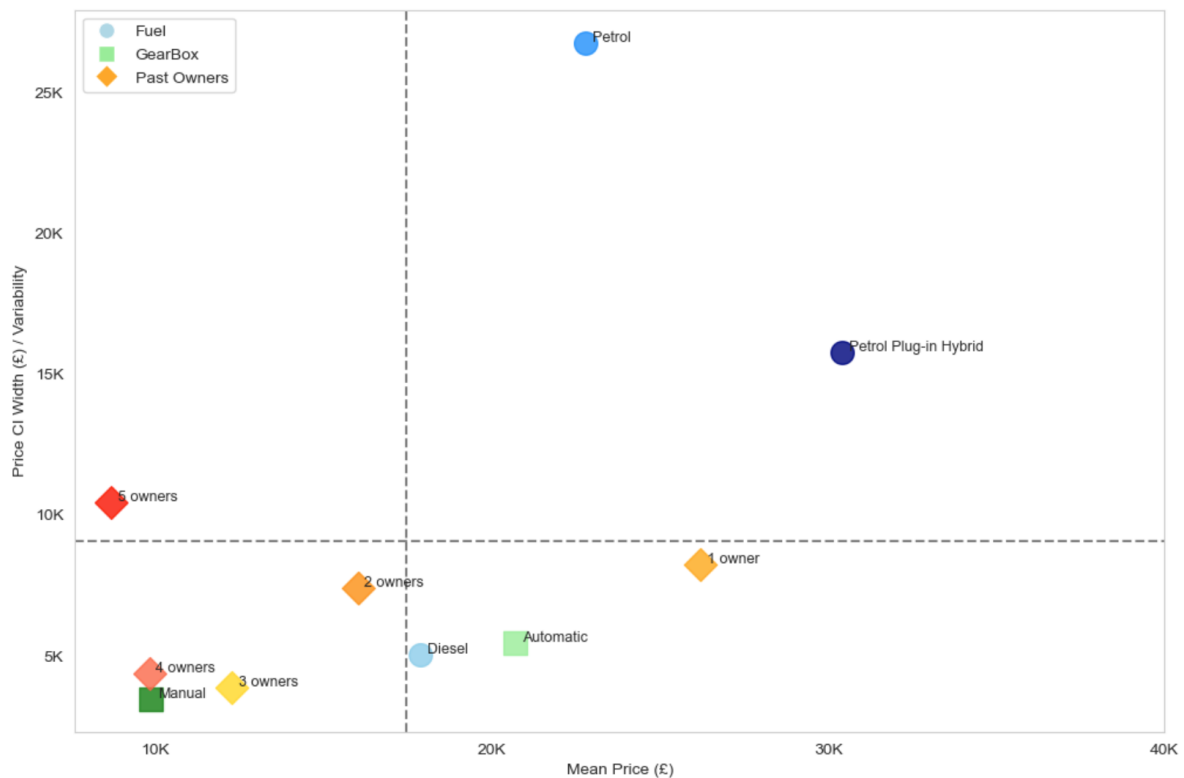
Variable (Factor)	df (Between, Within)	F-Statistic	p-Value	Significance ($\alpha=0.05$)	Interpretation
Past Owners	(6, 93)	7.00	0.000004	Highly Significant	Number of previous owners significantly affects price; cars with fewer owners retain higher resale value.

1.4.1.2 Paired T test Addressing Question 2: Mean price vary by the amount of time a listing stays on the website

Statistic	Value
Mean Price Difference (October 16 – November 05)	-277.04
t-Statistic	5.173
p-Value	0.0000

1.4.2 Precision of Estimates

All categories under Fuel, GearBox, and Past Owners (1–4) show a high likelihood of influencing the mean price of the **Range Rover Evoque**, with fewer past owners, automatic GearBox, and hybrid/petrol fuel types generally commanding higher and more stable prices.



The BCG-style matrix was designed using Gestalt principles of visual perception to ensure clarity and intuitive grouping. Categories were grouped and distinguished through proximity and similarity, with related vehicle types positioned close together and assigned consistent shapes or colors. Enclosure and connection helped visually differentiate clusters of fuel types, gearbox types, and past owner categories, while figure/ground made the data points stand out against the background for immediate focus. Subtle color gradients and marker variations emphasized differences in mean price and price variability, effectively conveying multivariate information in a single view.

By applying these principles, the matrix communicates patterns and relationships in pricing clearly and efficiently, making it easy for viewers to interpret trends and identify high-variability versus stable categories.

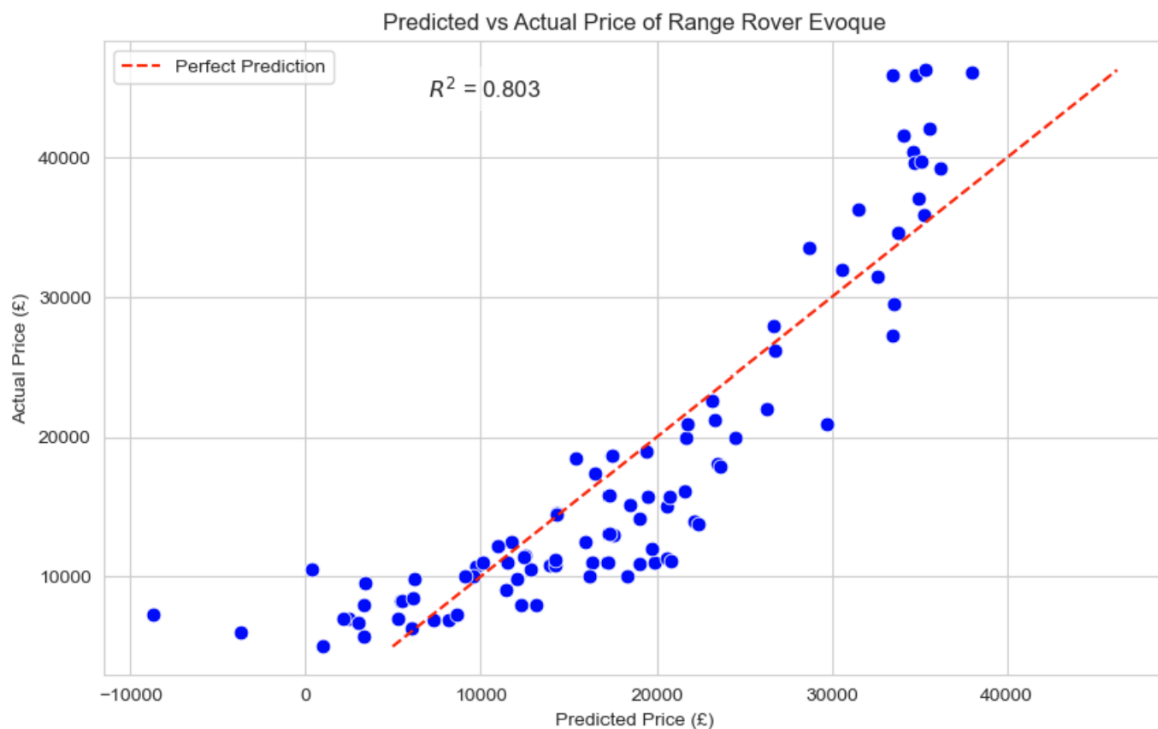
1.4.3 Determining the Key Price Drivers

Following the statistical tests we can conclude that the following variables are the key price drivers of Range Rover Evoque used cars.

- **Fuel Type**
- **Gear Box**
- **Past Owners**
- **Mileage**
- **Amount of time a listing is on the website**

1.4.4 Implementing Regression Analysis for Predictive Pricing

Question 3: Can we predict price (as collected on October 16) with the key price drivers as identified? (mileage, age, fuel, gearbox, past owners)? **Yes.**



Details in Appendix.

1.4.4.1 Prediction Model

Price=£37,950-£0.30×Mileage-£838×Past Owners-£833(if Petrol)+£2,116(if Petrol Plug-in Hybrid)-£3,113(if Manual)

Key Message:

The chart compares the predicted prices from our pricing model against the actual market prices. Points close to the red dashed line indicate accurate predictions, showing that the model can reliably estimate vehicle prices.

Intercept (37,950): Base price for a Diesel automatic car with 0 mileage and 0 past owners.

Mileage (-0.303): Price decreases by ~£0.30 per mile.

Past Owners (-838): Each additional past owner reduces price by ~£838 (on average).

Fuel adjustments:

- Petrol reduces price by ~£833 compared to Diesel.
- Petrol Plug-in Hybrid increases price by ~£2,116 compared to Diesel.

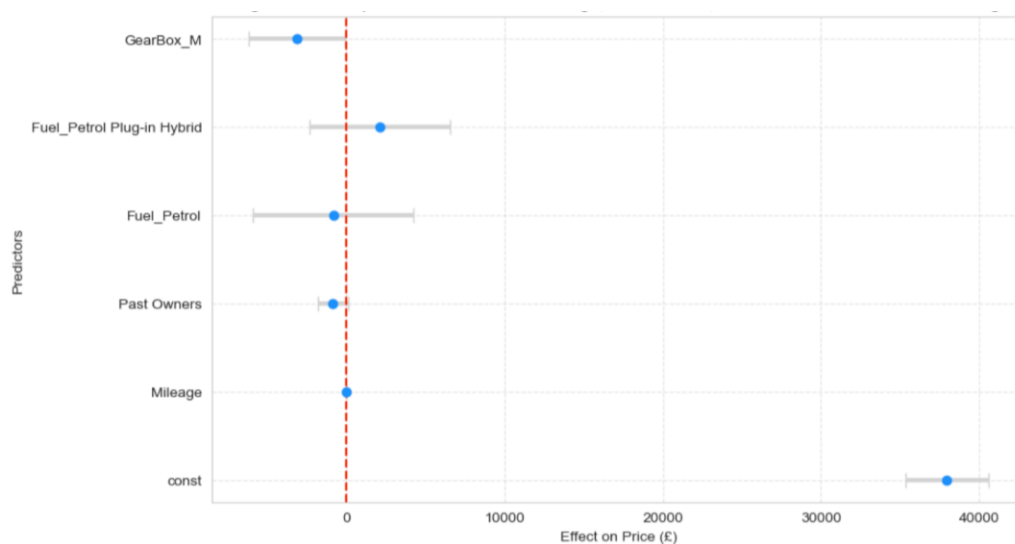
GearBox Manual (-3,113): Manual cars are ~£3,113 cheaper than Automatic cars, all else equal.

1.4.4.2 Validating Predictive Accuracy

To assess the reliability of the pricing model, we examined the estimated effect of each predictor along with its 95% confidence interval. The error bar chart shows how strongly each variable influences the predicted price and the uncertainty around each estimate.

- Predictors with confidence intervals that do not cross zero (e.g., Mileage and GearBox) are statistically significant, meaning we can be confident these features meaningfully impact price.
- Variables with intervals crossing zero (e.g., Fuel type, Past Owners) have less certain effects, indicating they are weaker predictors.

Effect of Predictors on Range Rover Evoque Price with 95% CI: Mileage, Manual Gear, and Number of Past Owners are Significant Predictor of Price



This validation ensures the model's predictions are robust and highlights which car features most reliably determine asking prices, providing the dealership with confidence for data-driven pricing decisions.

Conclusion and Suggestion Moving Forward

The analysis of used Range Rover Evoque prices around the Stratstone Reading dealership reveals clear patterns in pricing dynamics. Key features such as Mileage, GearBox type, and Fuel type significantly influence asking prices, while Past Owners shows a weaker effect. ANOVA and regression analyses confirm that mean prices differ across categories, with higher variability observed for hybrid fuel models and cars with 1–2 past owners, indicating less predictable pricing. Diesel SUVs and vehicles with 3–5 past owners have more stable prices, reducing risk for both buyers and sellers. The predictive pricing model explains over 80% of the variance in price, providing a reliable tool for forecasting asking prices and informing inventory and pricing strategies.

SUGGESTIONS FOR FUTURE REPORTS

- Expand the dataset to include additional vehicle features (e.g., service history, accident records) to improve predictive accuracy.
- Automate data collection to reduce manual errors and allow regular reporting.
- Track transactional prices alongside asking prices to better capture market realities.
- Explore additional modelling approaches (e.g., machine learning regression, time-series analysis) to account for non-linear relationships and market trends.
- Continue monitoring price variability across categories to refine pricing and promotional strategies, particularly for high-uncertainty segments like hybrid models and low-owner vehicles.

Appendix

Sample Data Set:

https://docs.google.com/spreadsheets/d/10NT9bUAsOs6DBFWODcCDqcEcOTN_s5U/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true

Population Data Set with Price Distribution Chart:

https://docs.google.com/spreadsheets/d/1yNzGu5WQYU3HX-u2S-R1eN08JclT-rSs/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true

Correlation Matrix: [https://docs.google.com/spreadsheets/d/16-](https://docs.google.com/spreadsheets/d/16-Gom9Jcoj4ecARryiP6Wyl5Cyr_aUpX/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true)

[Gom9Jcoj4ecARryiP6Wyl5Cyr_aUpX/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true](https://docs.google.com/spreadsheets/d/16-Gom9Jcoj4ecARryiP6Wyl5Cyr_aUpX/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true)

Box Plots: [https://docs.google.com/spreadsheets/d/1u8aMd4zUa5X8GhyDFalCxT5f6-](https://docs.google.com/spreadsheets/d/1u8aMd4zUa5X8GhyDFalCxT5f6-idFLEl/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true)

[idFLEl/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true](https://docs.google.com/spreadsheets/d/1u8aMd4zUa5X8GhyDFalCxT5f6-idFLEl/edit?usp=drive_link&ouid=106419493699029158184&rtpof=true&sd=true)

Owner vs Fuel Type Chart: [https://drive.google.com/file/d/1sfpGQl0Ig-w-](https://drive.google.com/file/d/1sfpGQl0Ig-w-VOKIEoJ6c_Rh8556sKi/view?usp=drive_link)

[VOKIEoJ6c_Rh8556sKi/view?usp=drive_link](https://drive.google.com/file/d/1sfpGQl0Ig-w-VOKIEoJ6c_Rh8556sKi/view?usp=drive_link)

Statistical Analysis, Precision of Estimate Chart, Regression and Predictive Accuracy Chart:

https://drive.google.com/file/d/1gq8K6x42BdMyknTTNctXutVooE7fTnQ8/view?usp=drive_li
[nk](#)